

S&DS 365 / 665
Intermediate Machine Learning

Sequence Models and Attention

November 17



Yale

For Today

Sequence models

- Memory circuits: GRUs (recap)
- Sequence-to-sequence models
- Attention mechanisms
- Transformers

But first the usual reminders

- Quiz 5 (last!) on Wednesday, Nov 19 (RL, HMMs, RNNs, GRUs)
 - ▶ No extensions
- Assn 4 due this Thursday, Nov. 20
- Assn 5 out; due Dec. 4
- Final exam: Friday, Dec 12, 9am
- Practice exams are posted
- Review sessions TBA

Recurrent neural networks

In an RNN, current output generated from a distributed state—a vector of neurons

- State is a deterministic, a nonlinear function of previous state and current input symbol.
- Dependence on earlier x_t 's is encoded in the state
- Output generated stochastically by sampling from

$$\text{Softmax}(\beta^T h_t)$$

where $h_t = f(x_1, \dots, x_{t-1})$ is the state vector.

“Vanilla” RNNs

In principle the state h_t can carry information from far in the past.

In practice, the gradients vanish (or explode) so this doesn't really happen

We need other mechanisms to “remember” information from far away and use it to predict future words

“Vanilla” RNNs

State is updated according to

$$h_t = \tanh(W_{hh}h_{t-1} + W_{hx}x_t + b_h)$$

We modify this with two types of “neural circuits” for storing information to be used downstream

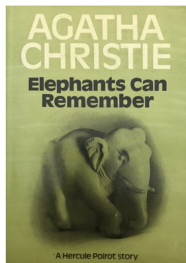
LSTMs and GRUs

Both LSTMs and GRUs have longer-range dependencies than vanilla RNNs.

We went through this in detail for GRUs, which are simpler, more efficient, and more commonly used

Problem 1 on Assn5

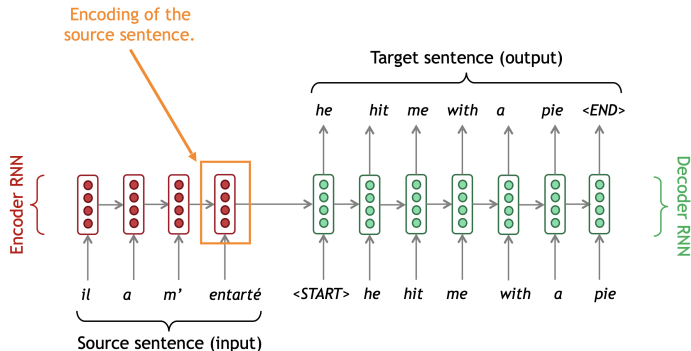
Problem 1: Elephants Can Remember (25 points)



In this problem, we will work with "vanilla" Recurrent Neural Networks (RNNs) and Recurrent Neural Networks with Gated Recurrent Units (GRUs). The models in this part of the assignment will be character-based models, trained on an extract of the book [Elephants Can Remember](#) by Agatha Christie. To reduce the size of our vocabulary, the text is pre-processed by converting the letters to lower case and removing numbers. The code below shows some information about our training and test set. All the necessary files for this problem are available through Canvas, under the file name "problem1_data".

Sequence-to-sequence models

- Important in translation
- Uses two RNNs (GRUs or LSTMs): Encoder and Decoder



"Sequence to sequence learning with neural networks," Sutskever et al. 2014, <https://papers.nips.cc/paper/5346-sequence-to-sequence-learning-with-neural-networks.pdf>. They also reversed the order of the input. Figure source: <http://web.stanford.edu/class/cs224n/slides/cs224n-2020-lecture08-nmt.pdf>

Sequence-to-sequence models

- The goal of Seq2seq is to estimate the conditional probability

$$p(y_1, \dots, y_T \mid x_1, \dots, x_S)$$

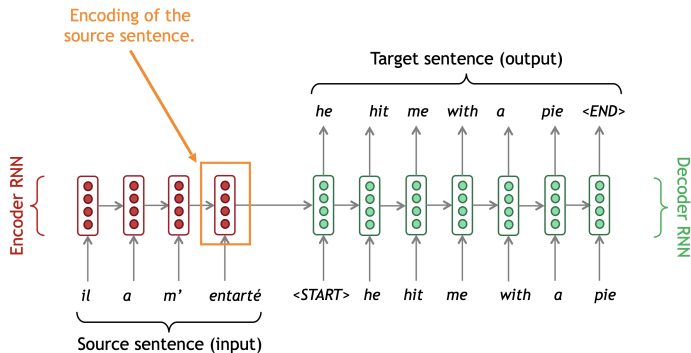
- Encoder RNN computes the fixed dimensional representation $h(x_{1:S})$; decoder RNN then computes

$$\prod_{t=1}^T p(y_t \mid h, y_1, \dots, y_{t-1})$$

- Original paper uses 4-layer LSTMs with 1000 neurons in each layer; trained for 10 days for a machine translation task.

Sequence-to-sequence models

This results in a “bottleneck” problem—all the information needs to be funneled through that final state.



Important modification: Attention

Attention



On each step of decoding, directly connect to the encoder, and focus on a particular part of the source sequence

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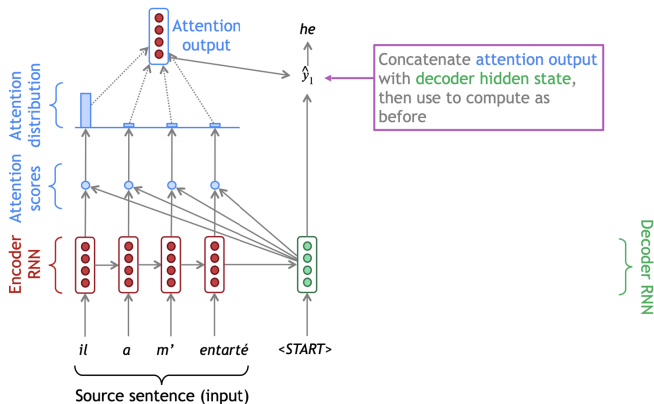


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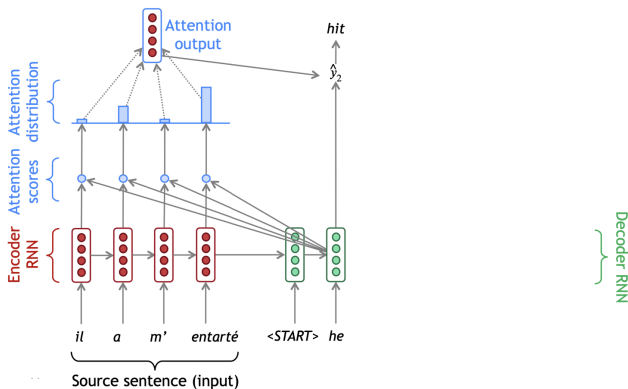


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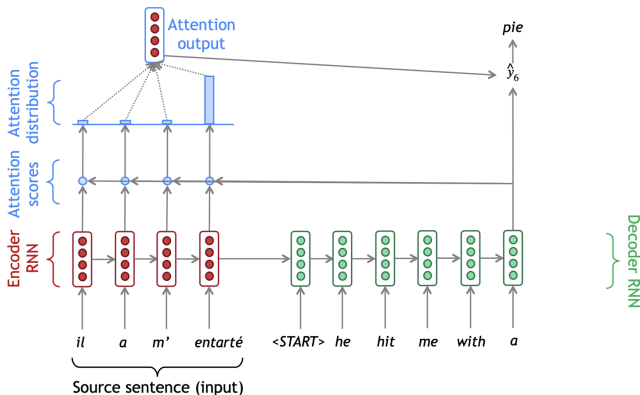


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Attention



On each step of decoding, directly connect to the encoder, and focus on a particular part of the source sequence

Modified RNN

A modified RNN model that incorporates this attention mechanism.

“Red states” r_t and “green states” g_t evolve according to Vanilla RNNs:

$$\begin{aligned}r_s &= \tanh(W_r r_{s-1} + W_x x_s) \\g_t &= \tanh(W_g g_{t-1} + W_y y_t) .\end{aligned}$$

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But target sentence depends on red states:

$$\begin{aligned}y_t &\sim \text{Softmax}(A g_t + B r_t) \\r_t &= \sum_{s \in \text{Source}} \alpha_{ts} r_s\end{aligned}$$

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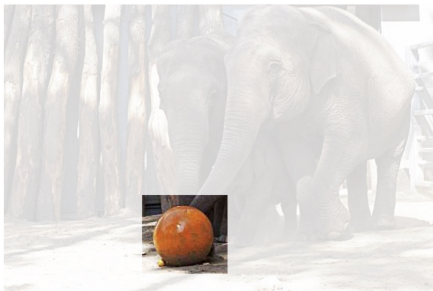
Attention weights α_{ts} depend on “similarity” between red and green states:

$$\alpha_{ts} = \text{Softmax}_s((Cg_t) \cdot (Dr_s))$$

Trainable parameters: A, B, C, D and W matrices



The two elephants played with an orange ball



The two elephants played with an orange ball



→ orange



→ ball

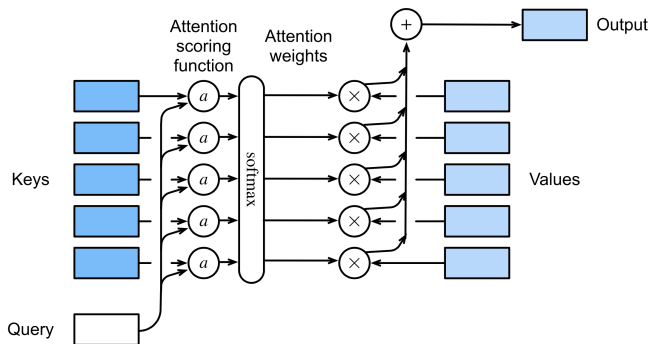
Attention in terms of query/key/values

Basic idea behind attention mechanisms:

- Neural networks so far: Hidden activations are linear combinations of input activations, followed by nonlinearity:
$$h = \varphi(Wu)$$
- A more flexible model:
 - ▶ We have a set of m feature vectors or values $V \in \mathbb{R}^{m \times v}$
 - ▶ The model dynamically chooses which to use
 - ▶ based on how similar a query vector $q \in \mathbb{R}^q$ is to a set of m keys $K \in \mathbb{R}^{m \times k}$.
 - ▶ If q is most similar to key i , then we use value (feature) v_i .

Attention in terms of query/key/values

Basic idea behind attention mechanisms:



Attention in terms of query/key/values

Basic idea behind attention mechanisms:

$$\begin{aligned}\text{Attn}(q, \{(k_1, v_1), \dots (k_m, v_m)\}) &= \text{Attn}(q, (k_{1:m}, v_{1:m})) \\ &= \sum_{i=1}^m w_i(q, k_{1:m}) v_i\end{aligned}$$

where weights w_i are softmax of attention scores $a(q, k)$:

$$w_i(q, k_{1:m}) = \frac{\exp(a(q, k_i))}{\sum_{j=1}^m \exp(a(q, k_j))}$$

Kernel regression as attention

Recall the kernel regression estimator:

$$\hat{m}(x) = \sum_{i=1}^n w(x, x_{1:n}) y_i$$

Here the attention scores (for Gaussian kernel) are

$$a(x, x_i) = -\frac{1}{2h^2} \|x - x_i\|^2$$

query: test point x

keys: data $x_1, \dots, x_n$

values: responses $y_1, \dots, y_n$

Dot product attention

Note that if x_i and x have a fixed norm then the attention scores are just scaled dot products:

$$a(x, x_i) = \frac{1}{h^2} x^T x_i$$

If query and keys have same dimension d , dot product attention is

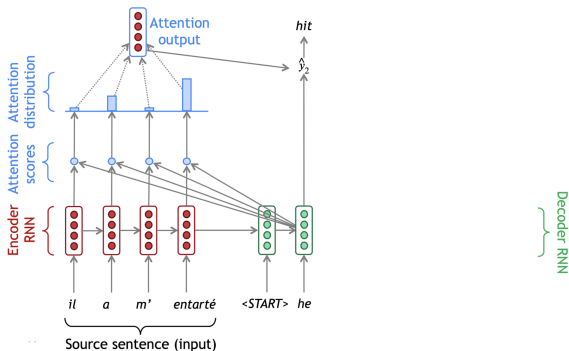
$$a(q, k) = q^T k / \sqrt{d} \in \mathbb{R}$$

and

$$\text{Attn}(Q, K, V) = \text{Softmax} \left(QK^T / \sqrt{d} \right) V$$

Retour d'entarté

In the *entarté* example, the keys and values are the red state vectors, the queries are the green state vectors



Key steps

- *Tokenize* the text into a fixed vocabulary
- *Embed* the tokens into high-dimensional vectors
- *Mix* the embeddings in the context using “Attention”
- *Map* the resulting vectors using a neural network

Additional components

- *Tokenize* the text into a fixed vocabulary
 - *Embed* the tokens into high-dimensional vectors
 - *Mix* the embeddings in the context using “Attention”
 - *Map* the resulting vectors using a neural network
-
- *Normalize* the embeddings
 - *Add* to residual stream

The vocabulary problem

- Choice of vocabulary is important
- We each know $\approx 40\text{K}$ words
- What if a word is not in the vocabulary?
- Need to be able to process—and generate—arbitrary text

Tokenization

Fugetsu-Do, a mochi store in the Little Tokyo neighborhood of Los Angeles, has existed for 121 years, with a lasting impact on its community.

Billy was a very baaaaaaaad goat.

This is another \$%^*ish example.

This is what we see

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Text

Token IDs

This is how the tokenizer groups bytes into tokens

Tokenization

```
[77845, 19209, 84, 12, 5519, 11, 264, 4647, 14946, 3637, 304, 279, 15013, 27286, 12818, 315, 9853, 12167, 11, 706, 25281, 369, 220, 7994, 1667, 11, 449, 264, 29869, 5536, 389, 1202, 4029, 382, 97003, 574, 264, 1633, 293, 29558, 33746, 329, 54392, 382, 2028, 374, 2500, 400, 46999, 9, 819, 3187, 382]
```

Text

Token IDs

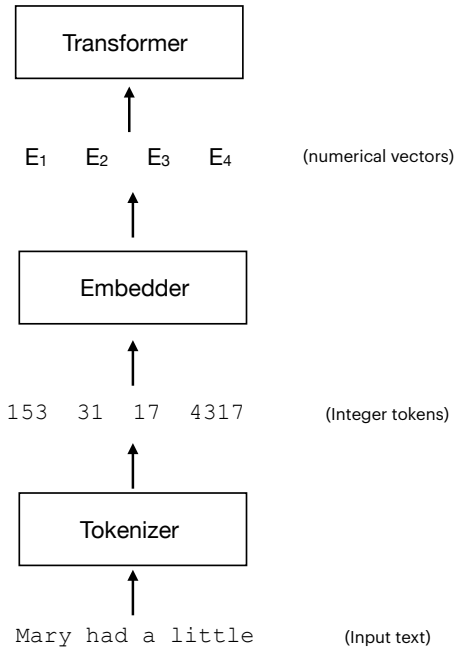
This is what the LLM sees—the “atoms” of an LLM

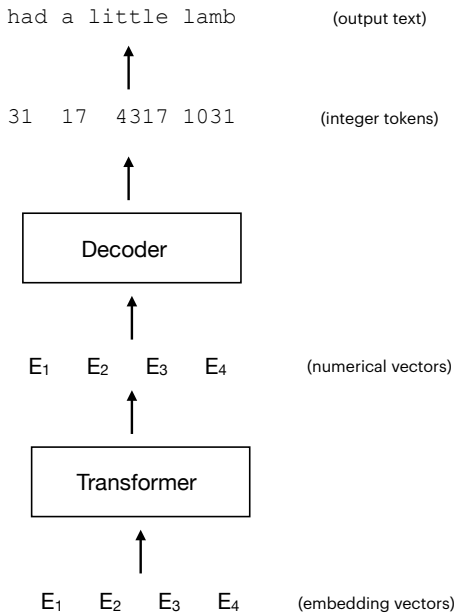
Tokenizer properties

- *Reversible*: Can convert from tokens back to original text
- *General*: Works on arbitrary text, even very different from training
- *Compressive*: Each token is about 4 bytes
- *Statistical*: Will break strings into frequently occurring pieces

BPE Algorithm – Byte Pair Encoding

- ① Input: Large sequence of bytes (UTF-8 encoding)
- ② Initialize: Each byte is a token; $V=256$
- ③ Iterate for K steps:
 - i Find pair of tokens (s, t) with largest count of co-occurring
 - ii Replace each occurrence of (s, t) with V
 - iii $V \leftarrow V + 1$
- ④ Final vocabulary size: $256 + K$





The key problem is integrating information across long distances

The leaves, as the weather turned cold in New Haven, silently from the in .

The key problem is integrating information across long distances

The leaves, as the weather turned cold in New Haven, fall silently from the trees in October.

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The leaves, as the weather turned cold in New Haven, fall silently from the trees in October.

To predict `fall` and `trees`, the model needs to refer back to `leaves` in memory (context):

- ▶ It's the subject of the sentence, and plural — syntax
- ▶ It has a “foliage” meaning — semantics

The leaves, as the weather turned cold in New Haven,

The leaves, as the weather turned cold in New Haven,

Intuition:

Embedding vector $\phi(\text{leaves})$ is added to the vector used to fill in the first blank.

Since $\phi(\text{leaves})$ is similar to $\phi(\text{fall})$, this boosts up the chances of generating `fall`

The leaves, as the weather turned cold in New Haven, fall silently from the

The leaves, as the weather turned cold in New Haven, fall silently from the

Intuition:

Embedding vectors $\phi(\text{leaves})$ and $\phi(\text{fall})$ are then added to the vector used to fill in the next blank.

This boosts up the chances of generating `trees`

Attention

- Which embeddings are added to current position is determined by *Attention*
- Attention is essentially a (very large) system of linear equations
- The weights in the linear system are optimized to extract the most predictive properties

Attention

- Which embeddings are added to current position is determined by *Attention*
- Attention is essentially a (very large) system of linear equations
- The weights in the linear system are optimized to extract the most predictive properties
- Attention is organized into multiple “heads”
 - ▶ Some may capture “syntax” (`subject` $\rightarrow$ `jverb`)
 - ▶ Some may capture “semantics” (`leaves` $\rightarrow$ `trees`)
 - ▶ But they are generally very difficult to interpret!

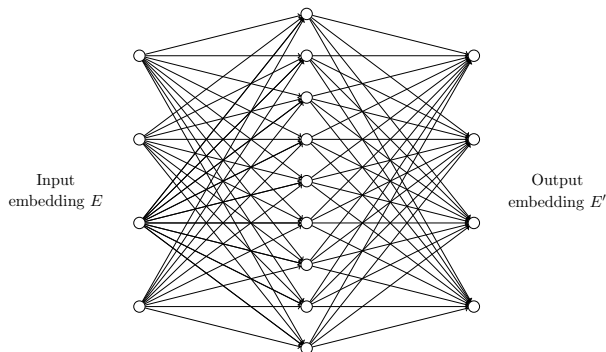
More intuition



- Different attention vectors are concatenated
- Capture different meanings of the sequence
- Words “hungry” and “sweet” are predicted using all of them

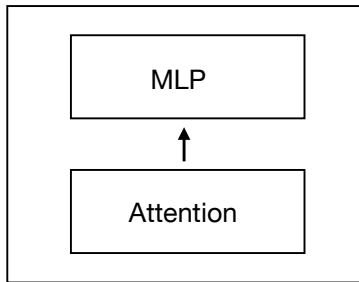
Adding nonlinearities

The last step is to take the resulting vectors (after they have been mixed up), and map them to new vectors using an MLP:

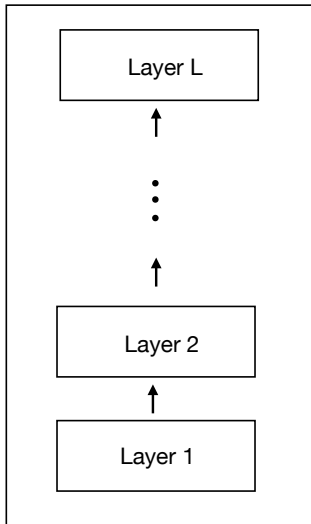


Creates new “features” to be used as embeddings in the next layer

Transformer Layer

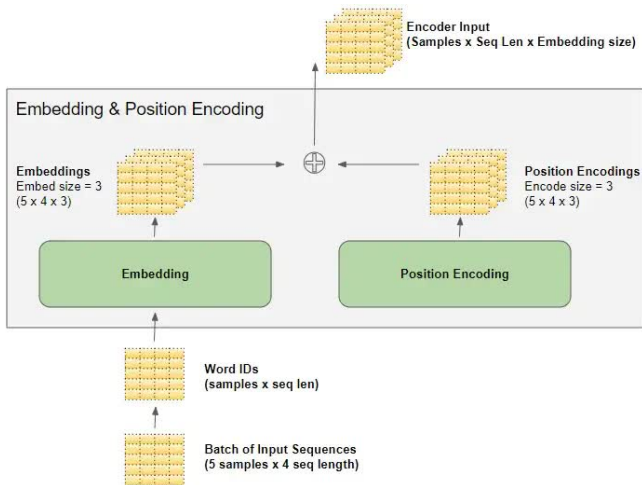


Transformer



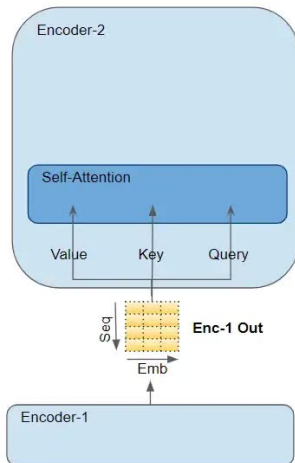
More detail on attention

Text processed in batches; embedding stage looks like this:



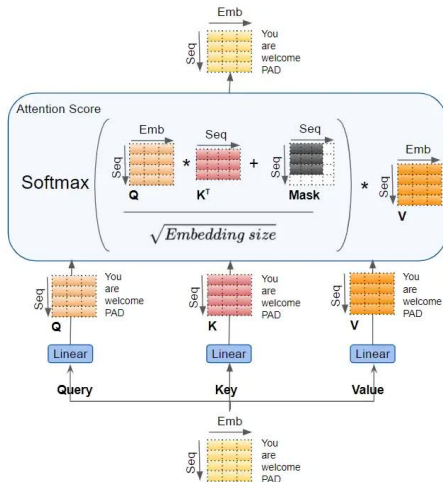
Multihead Attention

Detail of self-attention component

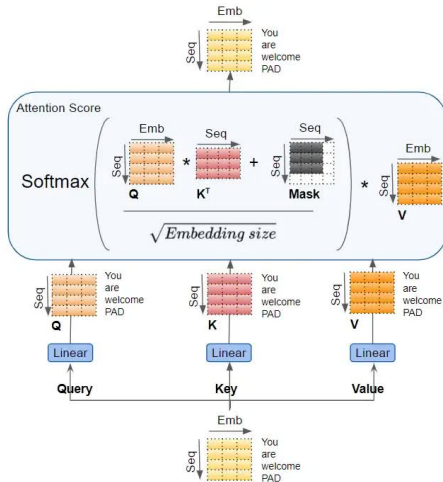


Multihead Attention

Detail of self-attention component (Blue are trainable parameters):

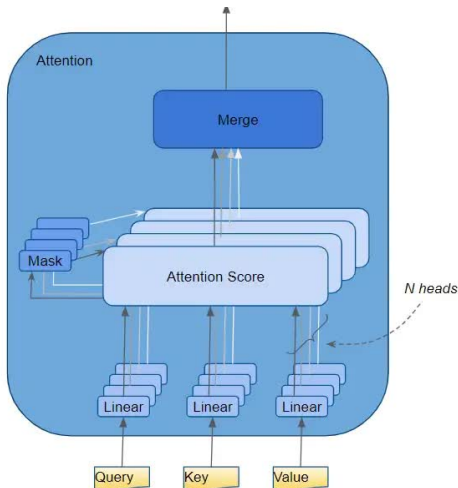


Multihead Attention



Mask is used in decoder to prevent use of future words

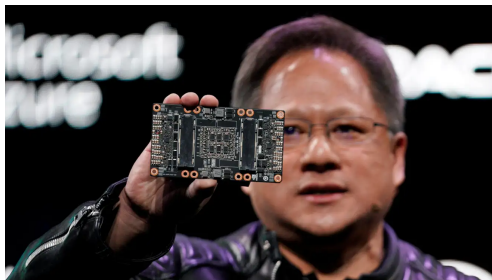
Multihead Attention



Multiple attention vectors are computed, then merged (concatenated)

Computation

- Everything is implemented as big matrix multiplies
- Carried out very efficiently on GPUs
- GPUs are optimized for linear equations (vector graphics)
- Allows scaling models to very large size



Back to intuition



- Different attention vectors are concatenated
- Capture different meanings of the sequence
- Words “hungry” and “sweet” are predicted using all of them

The residual stream

For *each* occurrence of a token in context, the GPT-3 Transformer maps it to 97 *different* embeddings (aka encodings)

- The first embedding represents the general token, across all contexts — 12,288 trainable numbers
- Each of the 96 layers then transforms this embedding using Attention and an MLP
- Result is $96 \times 12,288$ numbers for each token occurrence — more than one million total numbers (2 million bytes)
- Not *directly* interpretable by humans; but can be probed

The residual stream

The context length for GPT-3 is 2,048 tokens

- Total numbers for each context: $2,048 \times 1.1 \text{ million} \approx 2 \text{ billion}$
- The numbers for previous layer are added to the numbers for the next layer
- This is called the *residual stream*
- Number of parameters in model: 175 billion
 - ▶ We'll describe how to count parameters next

Transformer architecture

- 1 Input: sequence of tokens $t_1, \dots, t_n$
- 2 $x = \text{embed_tokens}(t)$
- 3 $x \leftarrow x + \text{positional_encoding}([n])$
- 4 Iterate for $\ell \in [L]$:
 - i $x \leftarrow x + \text{Attention}_\ell(x)$
 - ii $x \leftarrow \text{Normalize}_\ell(x)$
 - iii $x \leftarrow x + \text{MLP}_\ell(x)$
 - iv $x \leftarrow \text{Normalize}_\ell(x)$
- 5 Output: $y = \text{prediction_head}(x)$

Problem 2 on Assn5

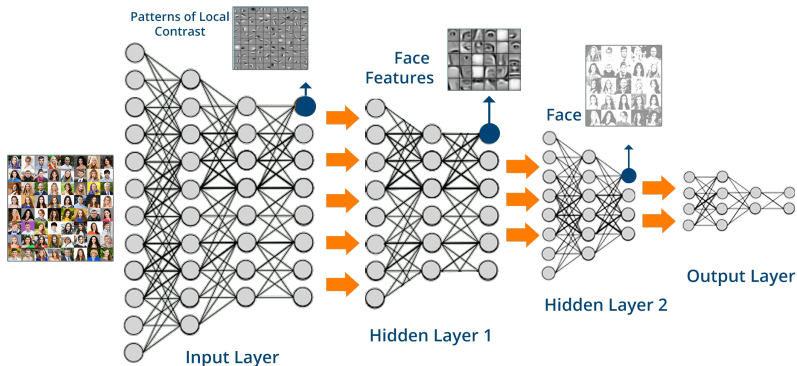
Problem 2. Be the Bard (35 points)



Transformer models are the current state of the art in many sequence modeling tasks, and the Transformer architecture underlies most Large Language Models (LLMs), including ChatGPT, Llama, Mistral, etc.

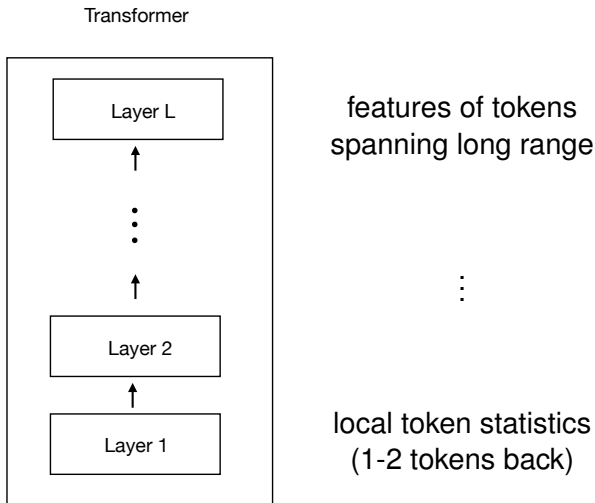
In this problem, we will implement a Transformer language model from scratch in `numpy`. You will be provided with the weights of a small Transformer language model that we trained on the works of Shakespeare. We will walk through implementing each component of the Transformer architecture and ultimately assemble this into a language model that can generate some text in the style of the Bard.

Increasing specificity of patterns



- Early layers capture local patterns of image contrast
- Later layers capture larger patterns that match parts of faces

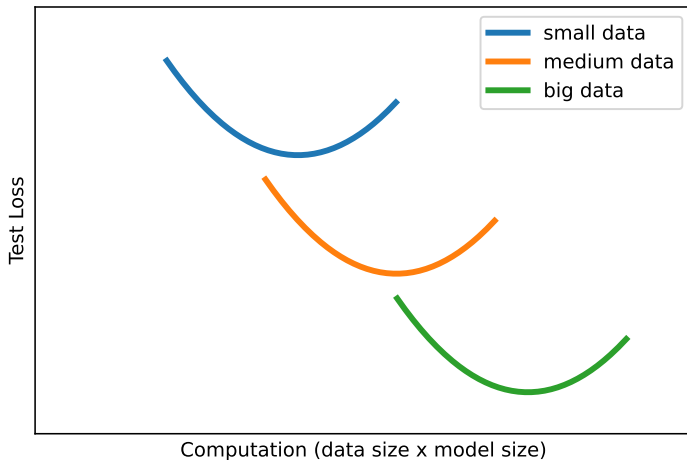
Increasing specificity of patterns



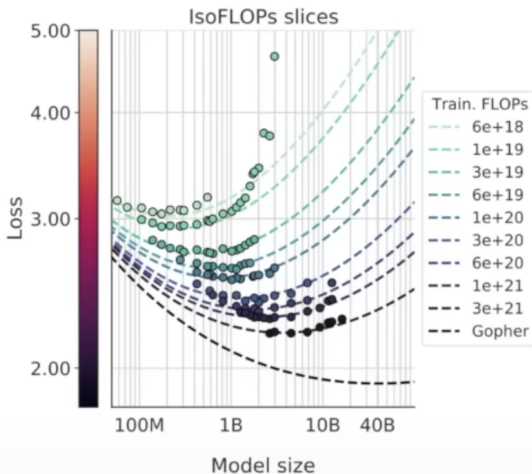
The Transformer architecture

How large are the models?

Recall: Scaling behavior of ML models



Recall: Scaling behavior of ML models



How does size of LLM scale?

Design choices: number of tokens V , dimension d of embeddings, and number of Transformer layers L

- Embeddings have $d \cdot V$ parameters
 - ▶ d numbers for each token, V tokens

How does size of LLM scale?

Design choices: number of tokens V , dimension d of embeddings, and number of Transformer layers L

- Self-Attention has $\approx 4d^2$ parameters
 - ▶ $d \times d$ matrix for queries (across heads)
 - ▶ $d \times d$ matrix for keys (across heads)
 - ▶ $d \times d$ matrix for values (across heads)
 - ▶ $d \times d$ matrix for combining heads

How does size of LLM scale?

Design choices: number of tokens V , dimension d of embeddings, and number of Transformer layers L

- MLP applied after has $\approx 8d^2$ parameters
 - ▶ Two layers
 - ▶ Hidden layer has $4 \cdot d$ neurons
 - ▶ Output layer has d neurons
 - ▶ $4d^2$ weights in each layer; $8d^2$ weights overall

How does size of LLM scale?

Design choices: number of tokens V , dimension d of embeddings, and number of Transformer layers L

- Embeddings have $d \cdot V$ parameters
- In each Transformer layer:
 - ▶ Attention has about $4d^2$ parameters
 - ▶ MLP has about $8d^2$ parameters
- Total trainable parameters: $12d^2 \cdot L + d \cdot V$

Size of models

GPT-2 (2019)

$$d = 1,600$$

$$V = 50,257$$

$$L = 48$$

$$12 \cdot d^2 \cdot L + d \cdot V \approx 1.5 \text{ billion parameters}$$

- ▶ Each GPT-2 Transformer layer has about 30 million parameters

Size of models

GPT-3 training data^{[1]:9}

Dataset	# tokens	Proportion within training
Common Crawl	410 billion	60%
WebText2	19 billion	22%
Books1	12 billion	8%
Books2	55 billion	8%
Wikipedia	3 billion	3%

Size of models

GPT-3 (2020)

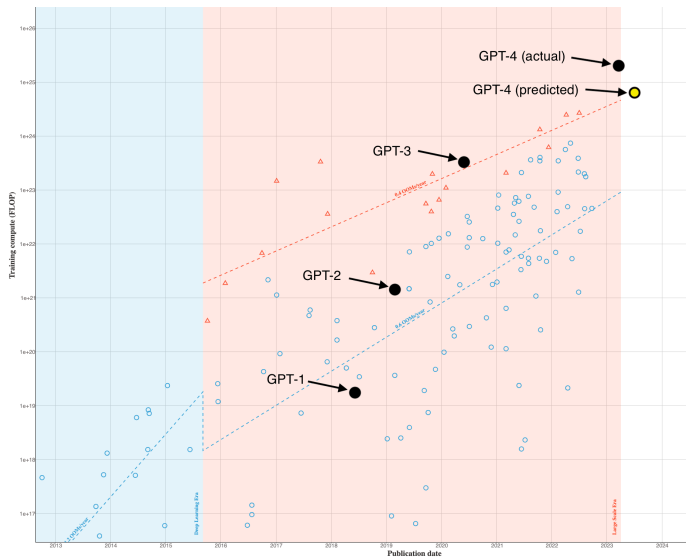
$$d = 12,288$$

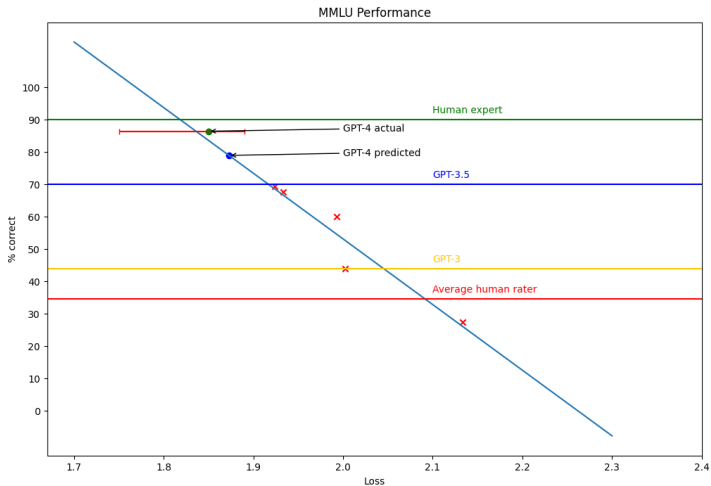
$$V = 50,257$$

$$L = 96$$

$$12 \cdot d^2 \cdot L + d \cdot V \approx 173 \text{ billion parameters}$$

- ▶ Each GPT-3 Transformer layer has about 1.8 billion parameters





MMLU: Massive Multitask Language Understanding, <https://en.wikipedia.org/wiki/MMLU>,
<https://paperswithcode.com/dataset/mmlu>

Sutton's "Bitter Lesson" (2019)



“The biggest lesson that can be read from 70 years of AI research is that general methods that leverage computation are ultimately the most effective, and by a large margin.”